M21267

Business Information Security Systems

14 February 2025

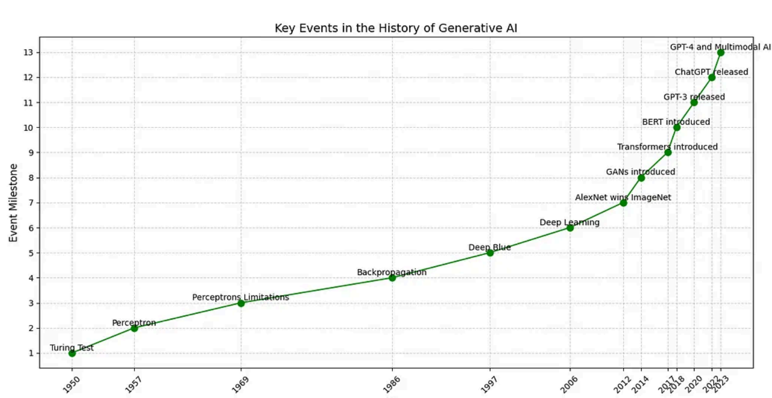
Reflective Report

Table of contents

Table of contents	**2**
Introduction	**3**
Task 1	**4**
Task 2	**5**
Task 3	**6**
Task 4	**7**
Task 5	**8**
Reflective Commentary	**10**

Introduction

Contrary to the belief that Generative Artificial Intelligence (GenAI) is a recent development, its origins trace back several decades. In the 1960s, Joseph Weizenbaum developed ELIZA, one of the first chatbots, which laid the groundwork for future advancements in GenAI. Over the years GenAI models have evolved significantly, leading to the sophisticated models we see today. GenAI now refers to AI systems capable of creating new content such as text, images, or audio by learning from vast datasets and generating outputs that resemble human-created material. Due to GenAI capabilities it can be significantly transforming for cybersecurity practices by enhancing threat detection, automating responses, and improving vulnerability management. By analyzing vast datasets, it can identify patterns indicative of potential security breaches, enabling proactive measures to prevent attacks. Additionally, it can aid the development of robust defense strategies by generating realistic attack simulations. For Small and Medium-sized Enterprises (SMEs), integrating GenAI into their cybersecurity strategies offers proactive defense against evolving cyber threats, even with limited resources. This report delves into the application of Gen|AI in cybersecurity, focusing on its relevance and benefits for SMEs.



Task 1

- I began task 1 by providing the assignment briefing to my GenAI that i will be using for this report. I decided to use one of the most commonly used AI, ChatGpt, and one that was released more recently, Qwen AI.
- The main difference between is that Qwwn excels at finding and citing reliable sources for research tasks, while ChatGpt is better at explaining information and providing a more natural, conversational experience
- As these tasks mainly focus on SMEs a set of five tasks has been given to test the capabilities, trustworthiness and reliability of different AI tools. Each task mainly focuses on aspects of cyber security.
- After providing the assignment brief to the tools to set up the foundation of my work. Using a specific prompt programmed both AI tools to act as Cyber Security experts while maintaining an understanding of me being a student. Instigated in my prompt that a list of references using APA7 formatting is provided as well a the end of each response to further validate the prompts outcomes.
- To prevent any unfair advantages to the AI models used I made sure to use the exact same prompts throughout.
- I began giving in the prompts of task 1 to provide a list of one major security breach for each year from 2020-2024 and provide reasoning.
- To make sure I got an accurate list I pushed both AIs to make sure these are 100% the most significant breaches.

Task 2

- My follow up was requesting it to analyze wisely from all the data breaches that were listed and proceed to wisely pick one of the breaches and give unique advice on how to prevent it.
- I emphasized the need for the advice to be SME relevant and not just generic.
- ChatGPT delivered stronger, more technical recommendations, including sandboxing file transfer tools, zero-trust segmentation, and using deceptive decoy files.
- I pushed both models to go further and refine the advice provided initially to make the output more actionable and industry aligned.

Task 3

- For this task a risk assessment matrix needed to be created. To guarantee that the results will be up to standard, i requested an analyzed and well structured table that includes risk assessment RAGing.
- The tables mainly consist of threats , description , likelihood , impact , overall risk assessment and mitigation strategy.
- ChatGPT provided a better understanding of the request as the table provided by ChatGpt was more advanced and detailed while the table provided by Qwen AI was a basic result and didnt go into too much detail.
- I refined the matrix by issuing a second prompt asking the model to critique and improve the initial output based on MITRE ATT&CK and CISA Frameworks that were also referenced at the end of the output.

Task 4

- For task 4 we are testing the AIs capabilities to create visual content. I instructed both AI models to create a well structured multimedia resource for an awareness campaign targeting the MoveIt breach.
- Both AIs began by giving me sets of instructions to create the requested poster, but to get the required outcome of actual poster i requested it to turn it into a pdf poster which still didnt give me the desired outcome from either AI.
- Finally asked it to make a nice image poster thats publishable which gave me the feedback I wanted. Although , Qwen AI showed its true limitations as it is unable to create any pdf images of any sort.
- Qwen AI provided different sorts of solutions to be able to fix its shortcomings such as different forms of AI that can generate the requested images or give me accurate prompts to place into different AI tools such as DALL.E , Canva or Lucidchart.
- This is particularly useful as it taught me how powerful visual communication is in cybersecurity. Traditional training documents often fail to engage users, but this visual resource could be printed, emailed, or used in onboarding packs. GenAI enabled me to combine story telling, layout design, and security expertise into unified asset.

Task 5

- Task 5 was set up as a way to challenge conventional formats and get different output for each exercise.
- To make sure I completely understand the task given I used the actual task as my initial prompt to explain and verify my understanding of the question
- I needed to recreate a different output for Task 2, 3, and 4.
- For Task 2 I revisited the advice and transformed the cyber security advice into a poem. The goal was to present technical ideas - like sandboxing, zero trust, honeypots, and patching discipline - in an emotionally engaging and metaphorical way.
- This showed me how GenAI can help make abstract or technical concepts meaningful to non-specialist audiences. A poetic format might be memorable to staff during cybersecurity awareness month or included included in newsletter to make security part of workplace culture.
- Furthermore, for task 3 i reimagined the matrix as an interactive board game. The board game created a visual breach timeline to simulate how the MoveIt attack unfolded. This transformation demonstrated that gamification could be a powerful teaching strategy.
- Finally, I developed a visual narrative showing the timeline of the MOVEit breach - from exploit discovery to exfiltration and response - alongside how it could have been prevented. This helped me think about how visual timelines can be

used in real-world incident response documentation.

GenAI helped push the boundaries of how cybersecurity advice can be designed, taught, and internalized. This task proved that AI isn't just about automation - it's a co-creator of educational tools and engagement strategies..

Reflective Commentary

This series of tasks provided a rich testbed for exploring GenAIs capabilities, and throughout, I adopted a reflective, iterative practice to improve my use of these tools. In this section, I discuss my overall experience using GenAI (especially ChatGPT and Qwen) across all tasks, comparing their performance, and highlighting lessons learned in prompting, verification, and creativity. I also reflect on how using GenAI influenced my decision-making with a focus on SME applicability.

- ❖ Evaluation of Prompting Skills:
  - ➢ Initially I started by approaching both GenAIs with building a foundation for the work we have to do. Using role prompting helped in getting the AI to research and find information in the exact field required without sidetracking.
  - ➢ I began by asking straightforward questions (e.g, "list one major breach from 2020-2024"), output was decent and direct but realized conversational prompting was for more powerful. Giving the AI detailed and specific instructions allows the outcomes to end up more accurate and prepared.
  - ➢ Being specific with the format was crucial for example: the matrix in task 3 explicitly requesting a table wit h certain columns was crucial.
  - ➢ A key prompting breakthrough was asking the model to explain its reasoning or critique its own answers. For example, having ChatGPT

double-check its risk ratings by explaining them led it to catch a couple of inconsistencies in its matrix.

- I also leveraged follow up prompts extensively: each GenAI output became an initial draft that i would always request to refine by requesting different formats. This iterative refining is something the literature calls prompt chaining, and experiencing it hands on showed me how responsive the models can be when guided step-by-step.
- My prompting became more efficient over time- by Task 5, I could quickly get the models into the desired creative mindset with a single well-crafted prompt, which earlier might have taken several attempts.

Model Comparison: ChatGPT vs Qwen

Using two different GenAI models in parallel was enlightening. ChatGPT (4o) generally produced more fluent, coherent, and detailed text. It was excellent at creative tasks (the poem, analogies, narrative flow) and explaining concepts in an accessible way. However, it had a tendency to hallucinate - it was caught investing statistics without backing it by a source. This reinforced the risk that generative models can output erroneous yet credible - sounding information.

Qwen on the other hand, acted almost like an academic assistant: it consistently provided citations and references for the outputs provided. It seems Qwen might be designed to retrieve information or was fine-tuned on knowledge bases, because it frequently cited CISA or MITRE content for justification. An example was in Task 1 where Qwen cited CISA alert for SolarWinds, which i verified as accurate. The trade off was that Qwens prose was more terse and at times disjointed; it also struggled with creative tasks such as creating media posters in any way (in addition, gave very generic suggestions for the board games, whereas ChatGPT dove right in). In practice, i found a complementary workflow: use ChatGPT for drafting and elaboration, use Qwen to fact-check and source information. When ChatGPT gave me a list of breaches, i cross-checked date and details via Qwen.

When Qwen gave a dry list of advice, i used the same prompt for ChatGPT and got more enriched answers with context and examples. Comparing the two also taught me about model bias: ChatGPT, with its training on internet text, sometimes gave widely discussed examples (it picked Nvidia Data breach for 2022 as expected), whereas Qwen occasionally surfaced less obvious answers (it mentioned a 2024 breach in a sector i had not considered until i merged our answers). For an SME user, this suggests that using multiple AI tools can yield more robust results - one model might catch what another misses or balance each others style (ones strength in explanation vs another in precision).

Strengths and Weaknesses of GenAI Observed

One clear strength is speed and breadth. GenAI dramatically accelerated research and drafting. Task 1s breach summaries or Task 2s recommendations would have taken me many hours to compile manually(digging through articles, writing summaries).With AI, I had initial answers in seconds and refined versions within minutes. Another strength is adaptability: the same tool that wrote a professional report could switch to creating a poem or game rules, which is astonishing. This adaptability meant I could approach a problem from multiple angles quickly (something that would be hard without a large team of specialists). On the flip side, a critical weakness is accuracy and trust. I learned to never accept an AI output at face value without verification. For instance, when asking the AI to provide references for each task, I had to manually verify each reference to prevent the presence of fabricated references or citations. This particularly taught me the importance of human oversight: the GenAI can draft, but a human must validate.

Another weakness is that GenAI sometimes lacks context awareness over long sessions. In Task 4, I had to further refine my prompt multiple times as the AI was unable to understand what i was requesting from it. This suggests that for lengthy or complex project, one must manage the context window carefully or break the task into smaller segments(which i did by resetting the conversation for each major artifact).

What Surprised me

I was consistently surprised by the creativity the AI exhibited. I went in thinking these models mainly regurgitate existing text, but the way ChatGPT crafted analogies or the board game concept felt genuinely innovative, as if I brainstorming with a colleague. I did not expect an AI to propose a fully structured game with roles and rules on the fly. Another surprise was how much AI “knew” about cybersecurity practices. For example, without me explicitly mentioning, ChatGPT in Task 2 brought up “zero trust” - a rather specific architecture principle. This indicates the training data included a lot of cybersecurity knowledge, which is encouraging for using GenAI in this domain. I also found the tone control impressive: I could instruct the AI to be more formal or more casual and it would rewrite content accordingly, which is valuable when preparing material for different audiences (executives vs developers vs general staff). What I struggles with at times was controlling the length of the output. Sometimes the GenAI models would produce very long answers when I needed concise ones. I learned the need to be more specific and using prompts such as “3-4 bullet points” or “keep it under 50 words”. Its a bit of an art to get the right verbosity.

Improvement Over Time

By task 5, I adopted a structured approach: planning prompts for specific goals (e.g., "focus on emotion" for a poem). Debugging outputs also improved - generic questions yielded generic answers, so specificity was key. External validation became routine, with cross-referencing via Qwen or manual searches to verify facts.

For SME contexts, I learned the necessity of adjusting prompts to reflect resource constraints (e.g., "50-person firm with no security team"). This redirected AI from enterprise-level suggestions (e.g., Security Operations Centers) to practical priorities like patching and backups.

Supporting SME-Oriented Decisions

GenAI aided decision-making by outlining options. For mitigations (Task 2), it highlighted cost-effective measures like patching and access control. However, recommendations required critical evaluation - e.g., dismissing “honeypots” as impractical for SMEs. AI broademed awareness of tools (e.g., Lucidchart) and frameworks (e.g.,NIST), but human judgement ensued relevance.

Overall Reflection

GenAI is a powerful ally but not a replacement for expertise. It excels at overcoming " blank page syndrome" and accelerating research, yet outputs demand validation. Transparency and documentation - such as noting AI-generated content - are crucial. For SMEs, combining models (e.g., self-hosted for privacy, public for general knowledge) balances strengths.

Social, Ethical, and Professional Issues

- Hallucination and Misinformation:
  - Fabricated refrences (e.g., ChatGPTs fake citation) underscore the need for independent verification, especially in cybersecurity where errors can be costly.
- Data Privacy:
  - Cloud-based models risk exposing sensitive inputs. Avoiding confidential data in prompts and considering on-premise solution.
- Human Oversight:
  - AI is a tool, not an autonomous decision-maker. The EU AI Act emphasizes human-in-the-loop for accountability.
- Accountability:
  - Professionals remain liable for AI-assisted outputs, Traceability (e.g., chat logs) aligns with due diligence.
- Bias and Fairness:
  - Models may reflect training data biases (e.g., Western-centric threats). Vigilance ensures inclusive outputs.
- Professional Integrity:
  - Transparency about AI use - without misrepresentation contributions - upholds ethical standards (e.g., ACM Code of ethics).